Restaurant Database Description

Overview

This restaurant database was created from the Yelp Academic Dataset, containing restaurant information and customer reviews. The database was designed with a strategic sampling approach to create a diverse, representative dataset for testing the Restaurant Reservation ChatBot.

Data Source

- **Primary Dataset**: Yelp Academic Dataset (Business and Review files)
- **Initial Filtering**:
  - Extracted only businesses categorized as "Restaurants"
  - Retained only city-state pairs with at least 5 restaurants to ensure meaningful representation

Sampling Strategy

The final database was constructed in 3 steps to ensure diversity and challenge:

1. Duplicate City Names

Method: - Identified all city names that appear in multiple states these cities include: - Audubon (NJ|PA) - Brentwood (MO|TN) - Clayton (MO|NJ) - Lebanon (IL|TN) - Southampton (NJ|PA) - Springfield (TN|PA) - Included **ALL restaurants** (560) from these duplicate-name cities - This creates natural ambiguity scenarios (e.g., Springfield, PA vs. Springfield, TN)

2. Major Metropolitan Areas

Method: - Selected the **top 6 cities** by restaurant count these cities include: - Philadelphia PA - Tampa FL - Indianapolis IN - Nashville TN - Tucson AZ - New Orleans LA - Included **ALL restaurants** (18910) from these major metropolitan areas

3. "Needle in Haystack" Samples

Method: - Randomly sampled **100 restaurants** from remaining locations nationwide in various smaller cities and towns some examples to test for are: - Hart's Tavern Lahaska PA - Red Robin Gourmet Burgers and Brews Wesley Chapel FL - The Country Club Chesterfield MO - Beachwood BBQ & Burger Saint Petersburg FL - Also included **100 non-restaurant businesses** as noise - First Call Demo and Clean Outs Philadelphia PA - Ace's Upholstery Sparks NV - Nashboro Animal Hospital Nashville TN

Review Data Integration

Review Processing: - Sampled **up to 10 reviews per restaurant** - Each restaurant entry includes: - Review texts (list of up to 10 reviews) - Star ratings (corresponding ratings for each review) - Number of sampled reviews